

UNIVERSITY OF LIBERAL ARTS BANGLADESH

Department of Media Studies and Journalism

RESEARCH PROPOSAL

Writing Without Writing:

**Generative AI Substitution, the Performance Gap, and the Foreclosure of Epistemic
Development Among Bangladeshi Undergraduates**

Submitted by

Md. Miraz Hossain

Student ID: 233012010

BSS in Media Studies and Journalism

Instructor, Academic English I — Centre for Language Studies (CLS), ULAB

Submitted to

Iram Mehrin

Senior Lecturer

Center for Language Studies (CLS)

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Abstract

Most undergraduates now write with AI. The question that perception-based surveys keep failing to answer is not whether they use it, but what their independent writing looks like once they have. This study tests, at a single Bangladeshi private university, whether students who delegate the cognitive work of writing to AI still develop the skills writing is supposed to teach.

The design is pre-registered, mixed-methods, and within-semester. We recruit 200–250 ULAB undergraduates and collect four kinds of data from each consenting student: a 50-minute, no-tech in-class essay at the start and end of the semester (the diagnostic); a take-home essay completed without AI under a comparable time budget (the no-AI control); a take-home essay completed with declared AI assistance (the AI-assisted condition); and a survey on AI use patterns and writing self-efficacy. Semi-structured interviews, run by a researcher with no teaching link to the participants, follow the quantitative phase. The lexical, syntactic, and argumentative features of all three written conditions are compared within each student, which lets us separate the contribution of AI from the contribution of time and setting.

Our working hypothesis is that students who use AI as a substitute, rather than as a thinking partner, will show a wider and widening gap between what they can produce in the AI-assisted condition and what they can produce in the no-AI conditions. We expect that gap to be largest among students who came into university with the least prior writing instruction. We propose “epistemic foreclosure” as an interpretive lens for the version of this problem that shows up in contexts where the underlying writing competence had not been developed in the first place — closed off rather than eroded. We treat foreclosure as a way of reading findings, not as a hypothesis that can be tested against atrophy in a single-semester window.

The study is exploratory and the design is correlational. It cannot establish causality and cannot generalize beyond institutions resembling ULAB. What it can do is produce the first performance-based evidence on this question from a Bangladeshi sample, and a methodology other institutions can borrow.

1. Introduction and Problem Statement

In spring 2026, while teaching Academic English at ULAB’s Centre for Language Studies, I noticed something I could not square. Students were submitting essays that looked competent: clean grammar, organised paragraphs, an argument of sorts. In class, when I asked them to defend the same argument out loud, many could not. They could not say why a piece of evidence supported their claim, could not push back on an objection, could not extend their own idea past the version they had written down. The writing was performing competence the thinking did not seem to back up.

That gap might be one teacher’s anecdote. I do not think it is. I think it is a pattern, and I think it tracks the arrival of generative AI as a stand-in for the cognitive work of writing — figuring out what you think, picking evidence, defending against the obvious counterargument, revising when the counterargument is good. When AI does that work for the student, the essay survives. The learning the essay was supposed to produce may not. The assessment becomes hollow.

The scale of the shift is unusual. Sajadieh et al. (2026), in the Stanford AI Index 2026, report that generative AI reached 53 percent global population adoption within three years of public release — faster than personal computers or the internet — and that four in five U.S. high school and college students now use AI for school-related work. Comparable Bangladesh-specific figures are scarce in the peer-reviewed literature. What we do know, from BTRC subscriber statistics and recent a2i digital literacy reports, is that smartphone-based internet access among urban university-aged Bangladeshis is effectively universal, that free-tier access to ChatGPT and Gemini is unmetered, and that AI-detection infrastructure has not been deployed at scale in Bangladeshi university assessment.

AI is not arriving into a system where students already write well. Rahman, Hossain, and Uddin (2024), in a qualitative case study at a single Bangladeshi government college, document weak academic writing among the students they observed: limited vocabulary, weak organisation, thin critical reasoning. They attribute these patterns not to individual failure but to an exam-centred secondary school system that rarely teaches writing as a process. Their findings are not nationally representative, but they fit anecdotal accounts from across the Bangladeshi tertiary sector, including ULAB’s CLS intake screening.

If that picture is roughly right, then the standard “atrophy” framing in the AI-and-writing literature does not quite fit here. You cannot atrophy a capacity you never built. AI in Bangladesh may not be eroding writing skill so much as replacing the productive struggle through which writing skill normally develops. We call this epistemic foreclosure — the closing-off of a skill that had not yet emerged — and we treat it here as a lens for reading the evidence, not as a separately testable hypothesis. (We say more about why in section 4.3.)

Peer-reviewed empirical work on generative AI in Bangladeshi higher education is, at the moment, almost absent from the published literature. Most of what exists globally is Western, survey-based, and cross-sectional: useful for tracking adoption, less useful for asking what students can do on their own once AI is in the room. This study is an attempt at that second question, in one institution, with measurements that do not depend on self-report. Findings will apply to ULAB and to private English-medium universities resembling it. They should not be read as a national picture.

The study is the first of two related projects on AI-mediated cognition in young urban Bangladeshis. A companion working paper on algorithmic news feeds and misinformation

susceptibility among Dhaka news consumers is in early development and will be posted to OSF as it matures. The pairing is not a research programme; it is two studies that share a question.

2. Research Objectives

1. To measure the within-person gap between AI-assisted and independent writing among ULAB undergraduates, using objective linguistic and argumentative complexity metrics, and to assess whether this gap changes over the course of a semester.
2. To examine the patterns and self-reported modes of generative AI use in academic writing at ULAB, and to identify student-level and contextual variables associated with the size of the performance gap.
3. To develop evidence-based, context-specific recommendations for assessment design and pedagogical practice that are both theoretically grounded and institutionally feasible within ULAB's constraints.

3. Research Questions

Primary research questions

RQ1. What is the magnitude of the within-person performance gap between AI-assisted writing and independent writing (no-AI take-home and no-tech timed conditions) among ULAB undergraduates, as measured by linguistic complexity and argumentative quality proxies?

RQ2. Is the performance gap associated with change over the course of a semester, and is any observed change moderated by AI use intensity, prior writing instruction, or faculty?

RQ3. What self-reported modes of AI use (assistive scaffolding versus substitutive bypass, as operationally defined in section 6.4) are most strongly associated with a larger performance gap, and what student-level variables predict the mode of use?

Qualitative sub-questions

RQ4. How do students describe the cognitive experience of writing with and without AI assistance, and what aspects of the writing process do they report delegating, retaining, or developing?

RQ5. What do instructors observe regarding shifts in student writing quality, classroom argumentation, and epistemic confidence in the context of AI adoption, and what barriers do they identify to pedagogical response?

4. Literature Review

Keywords: Generative AI in higher education · essay-based assessment validity · epistemic development · cognitive offloading · desirable difficulty · performance gap · Bangladesh higher education · NLP text analysis

4.1 Writing as epistemic practice

Essay-based assessment rests on a body of scholarship that treats writing not just as the communication of existing knowledge but as a means of producing it. Bereiter and Scardamalia (1987) draw the foundational distinction between knowledge-telling and knowledge-transforming writing: transformative writing, in which the writer engages with ideas rather than transcribing them, is the mode associated with the development of higher-order reasoning. It is also the mode most vulnerable to displacement by AI systems that deliver polished text without the underlying cognitive engagement.

Flower and Hayes (1981) identify planning, translating, and reviewing as the three constitutive stages of writing, each involving active reasoning that can be bypassed when AI generates substantive content. Vygotsky (1978) frames productive struggle in writing as developmentally necessary scaffolding: the challenge through which competence is constructed. Bjork (1994) extends this with the concept of desirable difficulty — the features of a task that make it cognitively demanding in the short term are the same features that produce durable learning. Removing those difficulties, as AI tools may do, risks removing the mechanism of learning.

Lodge, Thompson, and Corrin (2023) frame the pedagogical challenge created by generative AI as an assessment validity problem. When AI-assisted work is submitted under assessment conditions that presuppose independent cognitive engagement, the assessment may no longer measure what it purports to measure. This validity question affects every essay submitted in an AI-accessible environment, regardless of whether individual students use AI.

4.2 Generative AI adoption and modes of use

Survey-based research in Western university contexts documents AI adoption rates of 60–85 percent among undergraduates, with a significant proportion reporting that AI generates substantial portions of written assignments (Chan & Hu, 2023). Chan and Hu's (2023) study of 399 students in Hong Kong is one of the most cited; it documents student perceptions of generative AI's benefits and risks but does not propose a formal typology of use modes.

We propose a working operational distinction, drawing on accounts from the cognitive offloading literature and from instructor-facing AI literacy frameworks (Vee, 2017, on coding-as-literacy and Lodge et al., 2023, on AI as assessment problem), between two modes of student AI use:

- *Assistive use*: AI is used for brainstorming, structure suggestions, grammar checking, or feedback on self-written drafts. The student remains the author of the substantive content.
- *Substitutive use*: AI generates substantive assignment content that the student edits lightly or not at all and submits as their own.

The boundary between these modes is not always cleanly perceived by students themselves, and lightly edited AI-generated text is frequently indistinguishable from student writing under standard assessment conditions. Perkins, Roe, Postma, McGaughran, and Hickerson (2024) provide the most rigorous available evidence on detection: the Turnitin AI detector flagged 91 percent of GPT-4-generated submissions as containing AI content, but identified only 54.8 percent of the content as AI-generated, and only 54.5 percent of staff-marked submissions were referred to academic misconduct review. Detection is therefore unreliable as a sole basis for assessment integrity.

Sajadieh et al. (2026) document that AI adoption correlates strongly with GDP per capita, which means lower-income countries are integrating AI-mediated information environments at a faster pace than their policy and literacy infrastructure can support. Bangladesh, as a rapidly digitising lower-middle-income society with effectively universal free AI access and nascent governance infrastructure, sits squarely at that intersection. Empirical research on generative AI in Bangladeshi higher education remains almost absent from the peer-reviewed literature, and findings from Western contexts may not transfer directly.

4.3 Cognitive offloading, desirable difficulty, and the foreclosure lens

The cognitive offloading literature provides a starting framework for concern about AI-assisted writing. Risko and Gilbert (2016) review experimental evidence on the mechanisms that trigger cognitive offloading and the short-run consequences for performance on the offloaded task. Their review does not establish that habitual offloading produces long-term reductions in independent capacity; that is an inference one might draw, but it is not what the review tests. We therefore treat the offloading framework as a source of theoretical motivation, not as direct evidence for the foreclosure account.

The standard “atrophy” framing common in the AI-and-writing literature creates a logical problem in under-resourced educational contexts: one cannot erode a capacity not yet established. Rahman et al. (2024) document that students at the Bangladeshi government college they studied arrive at tertiary education with substantial deficits in academic writing, attributable to an examination-oriented school culture. Their findings are local but plausible for the broader Bangladeshi context. Into this context, AI may not arrive to erode a competence but to foreclose the development of one.

Epistemic foreclosure, as we use the term, is a conceptual lens rather than a separately testable theoretical claim. Empirically, foreclosure and atrophy predict the same observed pattern: a within-person gap between AI-assisted and independent writing that is larger among heavier AI users. A single-semester cross-sectional comparison cannot distinguish the two developmental stories. We therefore use “foreclosure” as a way of reading findings in low-baseline contexts, not as a hypothesis pitched against atrophy. The empirical test the study does run is whether the predicted pattern is present at all, which is consistent with either account.

Kahneman (2011) provides a related framing: AI writing tools allow students to remain in System 1 (fast, intuitive, low-effort) processing, potentially deferring the development of System 2 (slow, deliberate, effortful) reasoning. The dual-process language is suggestive rather than directly operationalised here, since Kahneman’s framework is about judgment under uncertainty, not about writing development. The augmentation literature raises a genuine counter-account: AI, used as scaffolding, may support rather than substitute for System 2 engagement. The study is designed to detect both trajectories.

4.4 Measuring what matters: the case for performance-based evidence

The dominant methodology in the AI-and-writing literature is the cross-sectional self-report survey. While useful for mapping prevalence and attitudes, self-report cannot address the most pressing empirical question: is AI use associated with a measurable difference in what students can do on their own? No published peer-reviewed study, to our knowledge, provides a within-person performance comparison between AI-assisted and independent writing in a Bangladeshi sample. The closest available evidence is from Western contexts and is largely indirect.

The within-person design used here is methodologically stronger than self-report alone. By collecting AI-assisted, no-AI take-home, and no-tech timed writing from the same students at two time points, we can examine: the magnitude of the gap at baseline, whether the gap changes over a semester, and whether any change is moderated by AI use intensity, faculty, year of study, or prior writing instruction. The design is still correlational and quasi-experimental; it cannot establish causality, and the limitations are taken seriously in section 6.10.

Pre-registration of hypotheses and analysis plan on OSF before data collection provides a layer of methodological transparency, protecting against post-hoc narrative-fitting and signalling to journal reviewers that the central findings are confirmatory rather than exploratory.

4.5 The ULAB context and the broader South Asian research gap

The global literature on AI in higher education is concentrated in Western, English-language contexts. Empirical research on generative AI in South Asian higher education is largely absent

from peer-reviewed journals. Findings developed in resource-rich contexts may not map onto contexts with different baseline writing cultures, different access patterns, and different institutional constraints.

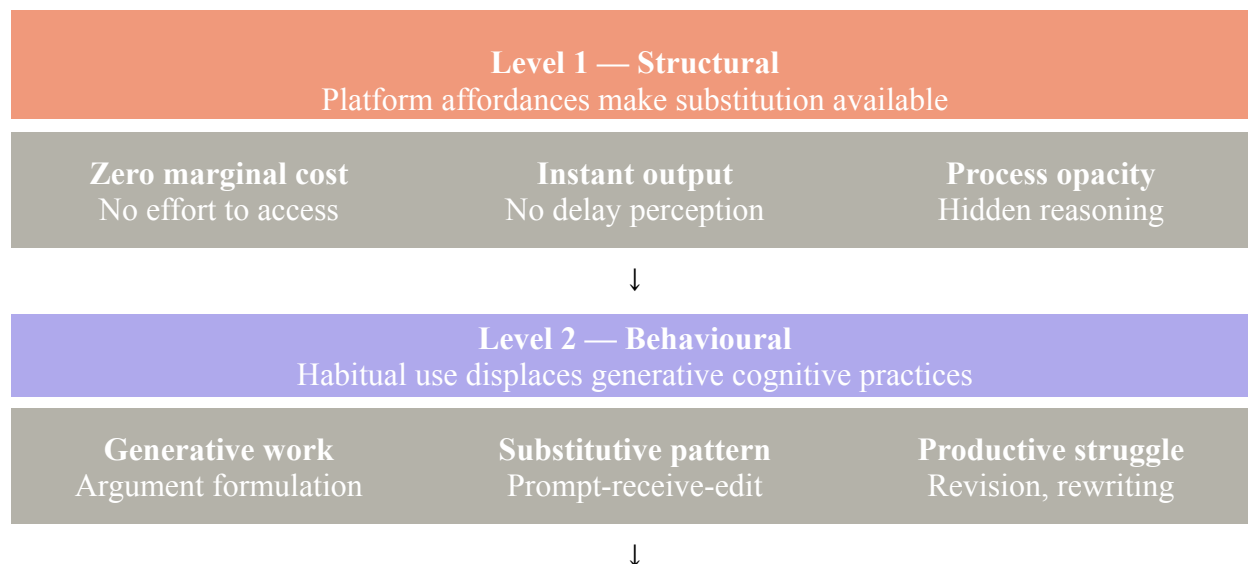
ULAB is a private, English-medium institution serving a predominantly urban, middle-class student population — a specific and atypical segment of Bangladeshi higher education. Findings from this study should not be extrapolated to Bangladesh’s public university system, the national university network, or Bangla-medium institutions without further research. What the study can plausibly offer is context-specific evidence for institutions sharing ULAB’s profile, and a methodological template that could be adapted for broader application. A nationally representative study would require substantially larger sampling across institution types and is beyond the scope of this project.

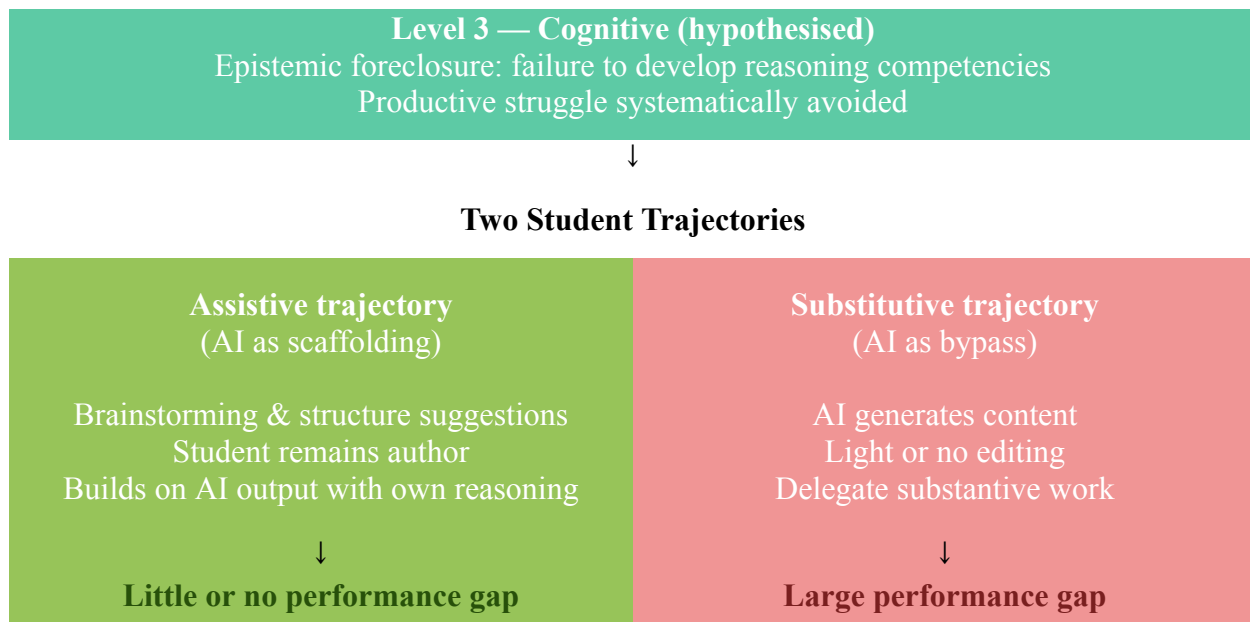
From a policy perspective, even institution-specific empirical data is a contribution, since ULAB administrators and educators are currently navigating AI adoption without any locally grounded evidence base. (For the related question of media literacy and information vulnerability in Bangladesh, see Muzykant, Hossain, Muqsith, and Fatima, 2022.)

5. Theoretical Framework

The framework proposes that AI substitution is associated with epistemic foreclosure through three interlocking mechanisms. The arrows in the framework represent hypothesised associations, not established causal chains.

AI Substitution → Epistemic Foreclosure: Three Linked Mechanisms





Level 1 — Structural. Platform affordances make zero-friction substitution available: AI tools offer zero marginal cost, instant output, process opacity, and output quality that typically passes inspection under standard assessment conditions. Unlike prior writing aids, AI can generate entire coherent essays at no cognitive cost to the student.

Level 2 — Behavioural. Habitual substitutive use may displace the generative cognitive practices of writing: original argument formulation, source search and evaluation, evidence chain construction, and revision under epistemic uncertainty. The dominant substitutive mode — prompt, receive, lightly edit, submit — constitutes a different cognitive activity from writing.

Level 3 — Cognitive (hypothesised). The cumulative effect of habitual substitution is hypothesised to be epistemic foreclosure: a failure to develop the reasoning competencies that writing is designed to cultivate, because the productive struggle that would have produced them is systematically avoided.

The framework distinguishes two student trajectories. Students who use AI as scaffolding (generating ideas or structures they then build on through their own reasoning) may show little or no performance gap. Students who use AI as bypass (delegating substantive content) are most likely to show a large one. The study design lets us operationalise and test this distinction.

The framework draws on Bjork (1994) on desirable difficulty, Vygotsky (1978) on the zone of proximal development, Bereiter and Scardamalia (1987) on knowledge transformation, and Kahneman (2011) on dual-process theory.

6. Methodology

6.1 Research design and pre-registration

The study uses a sequential explanatory mixed-methods design with a within-person quasi-experimental component. It has three phases:

- *Phase 1*: Objective performance measurement. Each consenting student produces writing in three conditions — a 50-minute no-tech in-class diagnostic, a take-home essay completed without AI under a comparable time budget, and a take-home essay completed with declared AI assistance. All three are submitted at the start and at the end of the semester.
- *Phase 2*: A quantitative survey on AI use patterns and academic writing self-efficacy.
- *Phase 3*: Semi-structured interviews with students and instructors, conducted by a third-party researcher with no teaching or evaluative link to the participants.

The third condition (no-AI take-home) was added to the design after early review feedback. It serves a specific methodological purpose: without it, a comparison between the no-tech diagnostic and the AI-assisted submission confounds the AI signal with differences in time pressure, revision opportunity, prompt format, and access to non-AI sources. The no-AI take-home matches the AI-assisted condition on all those dimensions except AI access. The clean within-person AI contrast is therefore between the no-AI and AI-assisted take-homes. The no-tech diagnostic provides a separate, independent measure of the student's writing under fully constrained conditions.

The study will be pre-registered on the Open Science Framework (OSF) before data collection begins. The pre-registration will specify all primary hypotheses, the operational definition of the performance gap (see 6.4), the statistical analysis plan including the numerical cut-points for AI use modes (see 6.4), and the criteria for evaluating the augmentation account (see 6.11).

6.2 Study setting and target population

The study will be conducted at ULAB, Dhaka. ULAB has been selected for three reasons. First, the lead researcher has direct and ethically appropriate access to the student population through existing academic roles. Second, ULAB's student population represents the demographic most likely to have adopted generative AI tools among Bangladeshi higher education institutions. Third, the university's Centre for Excellence in Teaching and Learning (CETL) provides an institutional home for research with direct pedagogical implications.

The target population is undergraduate students at ULAB across all faculties who are enrolled in at least one course with essay-based written assessment in the study semester. The sample is explicitly ULAB-specific; generalisations to Bangladesh's broader higher education sector are made cautiously and with appropriate caveats.

6.3 Recruitment

Recruitment is run by a research assistant outside the lead researcher's teaching reach, to reduce the implicit pressure that a teacher-as-recruiter would create. The sampling frame is the ULAB registrar's roster of students enrolled in courses with essay-based assessment in the study semester. From that frame, students are sampled with stratification by faculty (proportional allocation) and year of study (proportional allocation).

Recruitment proceeds in two waves. The first wave invites approximately 600 students by email and through in-class announcements in non-researcher classrooms; the target enrolment is 250 consenting students, allowing for an assumed 40 percent consent rate informed by comparable Bangladeshi university survey response rates. If first-wave consent falls short, a second wave samples additional names from the frame.

Participants receive a 500 BDT mobile recharge or equivalent on completion of the full data collection sequence, paid by the lead researcher's discretionary funds. Compensation is disclosed in the recruitment materials and the consent form.

6.4 Phase 1: Objective performance measurement

Condition A — No-tech in-class diagnostic

At the start of the study semester, each consenting participant completes a 50-minute no-tech essay of approximately 500 words on a prompt drawn from a pool calibrated to the year of study. Phones, laptops, and external materials are prohibited. The diagnostic is administered in a classroom run by an instructor outside the Centre for Language Studies, to remove any teacher-as-evaluator pressure from the lead researcher. The same exercise is repeated at the end of the semester with an alternate prompt drawn from the same calibrated pool.

Condition B — No-AI take-home essay

In the same window as conditions A and C, each participant completes a take-home essay of approximately 600 words on a prompt drawn from the same pool, with an explicit declaration: no AI of any kind, no large-language-model paraphraser, no AI-assisted grammar tools. Participants commit to this in writing and agree to a brief reflective interview at the end of the semester in which the no-AI commitment is discussed. We accept that compliance cannot be fully verified; the no-AI condition is treated as a participant-declared rather than verified state, and the analysis reports any indicators of non-compliance flagged by metric inconsistency.

Condition C — AI-assisted take-home essay

In the same window, each participant submits a take-home essay of approximately 600 words on a comparable prompt, with declared AI assistance and a brief description of what was prompted,

what was kept, and what was edited. We collect, with consent, both the final submission and the participant's brief use log.

The within-person AI signal is the contrast between conditions B and C. The no-tech diagnostic A provides an independent measure of constrained-condition writing performance.

NLP-based artefact analysis

All three conditions are analysed using Python-based NLP tools. Four metric dimensions are used, each justified by its relationship to the constructs of interest. We treat all metrics as proxies for argumentative and epistemic engagement, not as direct measures of cognitive capacity.

- *Lexical diversity*. Primary measure: Moving-Average Type-Token Ratio (MATTR), preferred over standard TTR because TTR is sensitive to text length and our conditions differ in length. Supplementary indicators: standard TTR (reported but not used for primary inference), proportion of academic vocabulary against the New Academic Word List, and proportion of rare vocabulary.
- *Syntactic complexity*. Mean sentence length, clausal density, and subordination index, computed using the spaCy dependency parser. These measures proxy the structural elaboration of argument. We acknowledge that higher syntactic complexity in AI-assisted text may reflect AI's averaging behaviour rather than the student's independent construction.
- *Argumentative structure*. Proportion of claim, evidence, and counterargument sentences, coded using the Argument Mining for Educational Use (AMEU) rule set adapted for the present prompts. The coding is piloted on 30 pre-AI-era ULAB student essays (from 2019–2021, drawn from the CLS archive) to establish a baseline distribution and inter-coder reliability. Target reliability is Cohen's $\kappa \geq .70$ on a 10 percent double-coded subset. We report κ in the final analysis. If reliability falls below target after a second training round, this metric is reported only descriptively, not used in inferential models.
- *Stylometric consistency*. Within-document stylistic variance computed from sentence-level embeddings (sentence-transformers; cosine distance between adjacent sentence embeddings, averaged within document). Abrupt stylistic shifts may indicate AI insertion. This metric is reported as a robustness check, not used as a primary outcome.

The lexical and syntactic measures are stylistic features. Their interpretation as proxies for epistemic engagement rests on the argumentative-structure measure carrying the central inferential load. The pre-AI-era pilot is the calibration step that lets us interpret deviations from baseline.

Operational definitions of AI use modes

Self-reported AI use intensity is measured as the percentage of the AI-assisted submission's content the student reports as generated, edited, or substantively rewritten by AI. For the purposes of analysis:

- *Substitutive use* is operationally defined as self-reported AI-generated content of more than 50 percent of the submission.
- *Assistive use* is operationally defined as self-reported AI-generated content of 50 percent or less.

These cut-points are pre-registered and treated as defensible operational choices rather than naturally occurring categories. Sensitivity analyses report the same models with cut-points at 30 percent and 70 percent. A secondary intensity measure is constructed from the participant's prompt-and-edit log (collected with condition C) and used as a triangulation check on self-report.

6.5 Phase 2: Quantitative survey

Survey instrument

The survey has five sections.

Section A collects demographic information.

Section B measures AI use patterns: tools used, frequency, mode of use (assistive vs substitutive on the operational definition above), and self-reported proportion of writing process personally performed.

Section C measures perceived effects of AI use on learning and reasoning, with items adapted from Chan and Hu (2023) and balanced to capture both foreclosure-consistent and augmentation-consistent effects. The augmentation-consistent items are not afterthoughts; the survey is designed to detect augmentation effects if they are present in the data, and the analysis plan treats augmentation findings as a primary deliverable, not an alternative to the primary one.

Section D measures academic writing self-efficacy. We use an adaptation of the Self-Efficacy for Writing Scale (SEWS; Bruning, Dempsey, Kauffman, McKim, & Zumbrunn, 2013), originally validated on middle- and high-school students. The adaptation extends item wording to university-level academic writing and is re-validated in the pilot phase (see 6.7) with confirmatory factor analysis on the pilot sample of 30 students. We report fit indices in the final paper and proceed with the adapted scale only if fit is acceptable ($CFI \geq .90$, $RMSEA \leq .08$).

Section E measures AI literacy: awareness of how AI language models generate text, understanding of their limitations, and capacity to critically evaluate AI output.

Power and sample

Target $N = 200\text{--}250$ students. A priori power analysis using G*Power 3.1 for the central moderation model (AI use intensity $\times$ prior writing instruction, predicting performance gap, with five covariates) at $\alpha = .05$ and power = .80 yields a minimum detectable f^2 of .04 at $N = 200$ and $f^2 = .03$ at $N = 250$ — i.e., the design can detect small interaction effects, but is underpowered for very small effects ($f^2 < .03$). If recruited N falls below 180, moderation findings are reported as exploratory, not confirmatory.

Analysis

Descriptive statistics characterise AI use patterns. Bivariate correlations examine associations between AI use intensity, mode of use, and performance gap. Multiple linear regression identifies variables associated with the performance gap while controlling for baseline diagnostic performance, year of study, faculty, and prior writing instruction. Moderation analysis (PROCESS macro Model 1) tests whether the association between AI use intensity and performance gap is moderated by AI literacy or prior writing instruction. Effect sizes (Cohen's d , partial η^2) are reported for all primary comparisons. The correlational nature of these associations is reported explicitly throughout; no causal inferences are drawn from regression results alone.

6.6 Phase 3: Qualitative interviews

Third-party interviewer protocol

All qualitative interviews are conducted by a third-party researcher with no instructional or evaluative relationship to the student participants. The interviewer is a research-trained colleague from outside CLS, briefed on the study framework and interview guide. The lead researcher does not see participant names or interview transcripts until thematic codes are finalised and identifiers are removed.

Sample and recruitment

Following quantitative analysis, 15–18 student participants are recruited using purposive sampling to ensure variation across performance gap size (large, moderate, small), AI use mode (substitutive vs assistive on the 50 percent cut-point), faculty, and year of study. Six to eight instructors from across ULAB departments are recruited for instructor interviews.

Interview protocol

Student interviews are semi-structured, approximately 45–60 minutes. The default language is English, since ULAB is English-medium and all participants have an English proficiency floor at

admission. Participants who prefer Bengali for sensitive material may switch at any point. Bengali segments are transcribed in Bengali, coded by a bilingual second coder against an English-translated transcript, and double-coded on a 20 percent subset for reliability. Member checks on translated quotes are offered to all participants.

The guide covers five domains: writing process and AI use patterns; what cognitive work is delegated versus retained; perceived effects on learning and reasoning; perceived change in independent writing capacity; and experiences of productive struggle.

Instructor interviews focus on observed changes in student writing and classroom argumentation, challenges in detecting AI substitution, and the feasibility of pedagogical responses within ULAB's constraints.

Analysis

Interviews are audio-recorded, transcribed verbatim, and analysed using reflexive thematic analysis (Braun & Clarke, 2006). The augmentation trajectory — cases where AI use appears to scaffold rather than substitute for epistemic development — is treated as a theoretically important deviant case and reported as a primary qualitative finding. Student and instructor data are analysed separately before integration.

6.7 Pilot phase

A 30-student pilot precedes full data collection. The pilot has three purposes: (1) re-validation of the SEWS-adapted self-efficacy scale via confirmatory factor analysis, (2) inter-coder reliability training on the argumentative-structure coding scheme, with target $\kappa \geq .70$ before the main analysis begins, and (3) testing of the no-AI take-home compliance procedure, including the brief reflective interview.

6.8 Integration of quantitative and qualitative data

Quantitative and qualitative findings are integrated at interpretation using a joint display and connecting strategy (Creswell & Plano Clark, 2018). The NLP and survey results establish what the performance gap pattern looks like and who is most affected; the qualitative findings explore how and why those patterns emerge from students' and instructors' accounts. Cases where a large quantitative performance gap is not accompanied by student-reported awareness of the gap are treated as analytically significant.

6.9 Ethical considerations

All participants receive a plain-language information sheet and provide written informed consent before data collection. Survey data is stored anonymously; interview participants receive

pseudonyms. No-tech diagnostic materials are linked to participants by code number only. The instructor-researcher power asymmetry is addressed structurally through three mechanisms: the third-party interviewer protocol, the diagnostic being administered in a classroom run by an instructor outside CLS, and explicit written assurance that no submission is shared with academic affairs, faculty, or grade records under any circumstance. Ethical approval is sought from the relevant ULAB faculty supervisor and the institutional ethics body before any data collection begins.

6.10 Limitations

This study has several important limitations that must be acknowledged.

Correlational design and confounds. The design is non-randomised and observational with respect to AI use. Students are not assigned to AI conditions; they self-select into patterns of use. The study cannot establish causality. The most significant alternative explanation is reverse causality: weaker writers may be more likely to adopt substitutive AI use, meaning any observed performance gap could reflect prior writing weakness rather than an effect of AI use. Controlling for baseline diagnostic performance in the regression models mitigates but does not eliminate this concern.

Metric limitations. The NLP metrics are proxies for epistemic engagement, not direct measures. MATTR, syntactic complexity, and argumentative structure coding are population-level patterns and are unreliable as individual-level AI detection tools. These metrics are used to characterise patterns across the sample, not to classify individual texts. Convergent validation against interview data is essential. The argumentative-structure metric carries the central inferential load; if reliability falls below the $\kappa \geq .70$ target, it is reported descriptively only.

Self-report and consent selection. The study cannot verify AI use from submissions alone. Students may under-report use; submissions may be partially AI-generated in ways the NLP analysis cannot reliably detect. The consent procedure itself biases the sample toward lower-intensity users; mitigations are described in 6.3 but cannot fully resolve the selection.

External validity. The sample is restricted to ULAB — private, English-medium, urban, fee-paying. Findings should not be generalised to Bangladesh’s public university sector, the national university system, or Bangla-medium institutions. The study is designed to provide institution-specific evidence and a replicable methodological template, not a nationally representative picture.

Time window. The 16-week semester window is too short to observe foreclosure as a developmental process. By the framework's own logic, foreclosure unfolds across years. This study can show the snapshot gap and any short-run change in the gap; it cannot show foreclosure developing. The study is best read as a cross-semester investigation that, if it finds the predicted pattern, motivates a multi-semester longitudinal follow-up.

No-AI compliance. The no-AI take-home condition relies on participant declaration; full verification is not possible. The analysis reports any indicators of non-compliance flagged by metric inconsistency, and treats the condition as participant-declared rather than verified.

6.11 The augmentation account as a primary deliverable

The augmentation account predicts that AI use, particularly assistive use, is associated with comparable or improved independent writing performance — a small or negligible performance gap, especially among students who use AI as scaffolding rather than bypass. The survey instrument, the qualitative interview guide, and the analysis plan all include provisions for detecting augmentation effects. If the data are more consistent with augmentation than with foreclosure, that finding is reported as the primary result. The framework section is written with the foreclosure case in mind because the proposal makes a case for a particular interpretive lens; the empirical analysis is written to give both accounts a fair test.

7. Workplan

Primary data collection is planned between June and November 2026, with preparatory and pre-registration work beginning in May 2026.

#	Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
1	Finalise instruments, ethics application, OSF pre-registration																
2	Ethics approval and informed consent protocols																
3	Pilot phase (N=30): SEWS re-validation, AMEU reliability training, no-AI compliance test																
4	Baseline no-tech diagnostic (Condition A)																
5	Baseline take-home essays (Conditions B and C)																
6	Full survey administration (200–250 students)																
7	NLP analysis of baseline artefacts																
8	Purposive interview recruitment (student + instructor)																
9	Semi-structured interviews (third-party interviewer)																
10	Interview transcription and translation																
11	End-of-semester diagnostic and take-home essays																
12	Performance gap analysis across the three conditions																

#	Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W1 0	W1 1	W1 2	W1 3	W1 4	W1 5	W1 6
1 3	Quantitative analysis (SPSS / Python)																
1 4	Qualitative thematic coding																
1 5	Write-up and discussion																
1 6	Supervisor review and revision																
1 7	Final draft and submission																

8. Expected Contributions

This study aims to make four contributions, with scope appropriate to a single-institution exploratory design.

First, the study will produce some of the first performance-based evidence on the relationship between generative AI substitution and independent writing capacity at the undergraduate level in Bangladesh. The within-person quasi-experimental design — including the no-AI take-home condition that isolates the AI signal from confounding task differences — is methodologically stronger than the perception-based studies that currently dominate the literature. Scope is limited to ULAB and comparable institutions.

Second, the concept of epistemic foreclosure is offered as an interpretive lens for AI-and-writing findings in contexts where foundational writing competencies are underdeveloped. We do not claim it is testable as a competing hypothesis against atrophy within this study's design; we claim it is a logically coherent reading of the evidence in low-baseline contexts that generates predictions that this study can examine.

Third, the study produces tiered, context-specific recommendations for ULAB, with the trigger-to-recommendation logic specified in advance: classroom-level interventions (process-based assessment components, in-class writing verification, AI-transparent pedagogy) follow from gap findings at the individual level; departmental-level interventions (assessment redesign policies, AI literacy modules) follow from gap findings clustered by faculty or year; institutional-level recommendations (AI governance frameworks) follow only if findings are robust across the full sample. The recommendation logic is published with the pre-registration so it cannot be retrofitted to the findings.

Fourth, the study contributes a methodological template that other Bangladeshi institutions can adapt: the three-condition writing protocol, the pre-AI-era calibration step, the third-party interviewer arrangement, the recruitment-bias mitigations, and the operational cut-points for AI use modes are documented so they can be reused or contested.

The study is offered as an exploratory but rigorous investigation that, if its core findings hold, would provide grounds for larger-scale longitudinal research and inform pedagogical responses at ULAB and comparable institutions. That is the scope. The contribution, within that scope, is real.

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